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Tectonophysics 343 (2001) 79–92

TECTONOPHYSICS

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Paleomagnetism in northwestern Bulgaria: geological implications of widespread remagnetization

Neli Jordanova [a,*], Bernard Henry [b], Diana Jordanova [a], Zivko Ivanov [c], Dimo Dimov [c], Françoise Bergerat [d]

[a] *Geophysical Institute, Bulgarian Academy of Sciences, Acad. G. Bonchev Str., block 3, 1113 Sofia, Bulgaria*
[b] *Paléomagnétism et Géomagnétisme, UMR 7577 CNRS and IPGP, 4 avenue de Neptune, 94107 Saint-Maur cedex, France*
[c] *Sofia University "St Kliment Ohridski", 15 Tzar Osvoboditel Blv., 1000 Sofia, Bulgaria*
[d] *Laboratoire de Tectonique, ESA 7072 CNRS-UPMC, B 129, P. et M. Curie University, 4 place Jussieu, 75252 Paris cedex 05, France*

Received 27 September 2000; accepted 13 September 2001

Abstract

Two different paleomagnetic directions have been obtained in several formations in northwestern Bulgaria. One of them, found only in Upper Permian–Lower Triassic red sandstones, is very likely a primary magnetization. A secondary magnetization of Eocene age appears as either pre-, syn- or post-folding depending on the site. The identification of the timing of remagnetization with respect to folding allows us to distinguish areas among the first deformed during Tertiary tectonics. Within each superimposed structural unit of the West Balkan, the deformation spread from south to north. The boundary between the Srednogorie and Balkan zones appears to be a major tectonic structure in Bulgaria. The Balkan likely underwent a clockwise rotation relative to stable Europe since the beginning of the main Middle Eocene orogenesis. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Paleomagnetism; Bulgaria; Remagnetization; Triassic

1. Introduction

Revealing the geodynamic history of southeastern Europe is important for a comprehensive understanding of the evolution of Peri-Tethys marginal structural units. To date, however, results from Bulgaria and Romania yield seemingly conflicting results. For example, northern Bulgaria has been considered in different studies either as an area that underwent no important rotation, or on the contrary, important counterclockwise rotation relative to stable Europe (Hsü, 1977; Sengör, 1984; Surmont et al., 1991; Bergerat et al., 1998; Muttoni et al., 2000), while data obtained in western Romania (Banat–Apuseni mountains) indicate a strong clockwise rotation (Surmont et al., 1990; Patrascu et al., 1990, 1992). Obtaining reliable well-dated paleomagnetic directions, constrained by stringent paleomagnetic tests, is key to resolving these discrepancies and understanding the structure of the eastern Carpathians and its relation to the stable part of Europe. The location of the study area is shown as a black rectangle on the sketch map in the left corner of Fig. 1.

* Corresponding author.
E-mail address: vanedi@geophys.bas.bg (N. Jordanova).

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PII: S0040-1951(01)00220-7

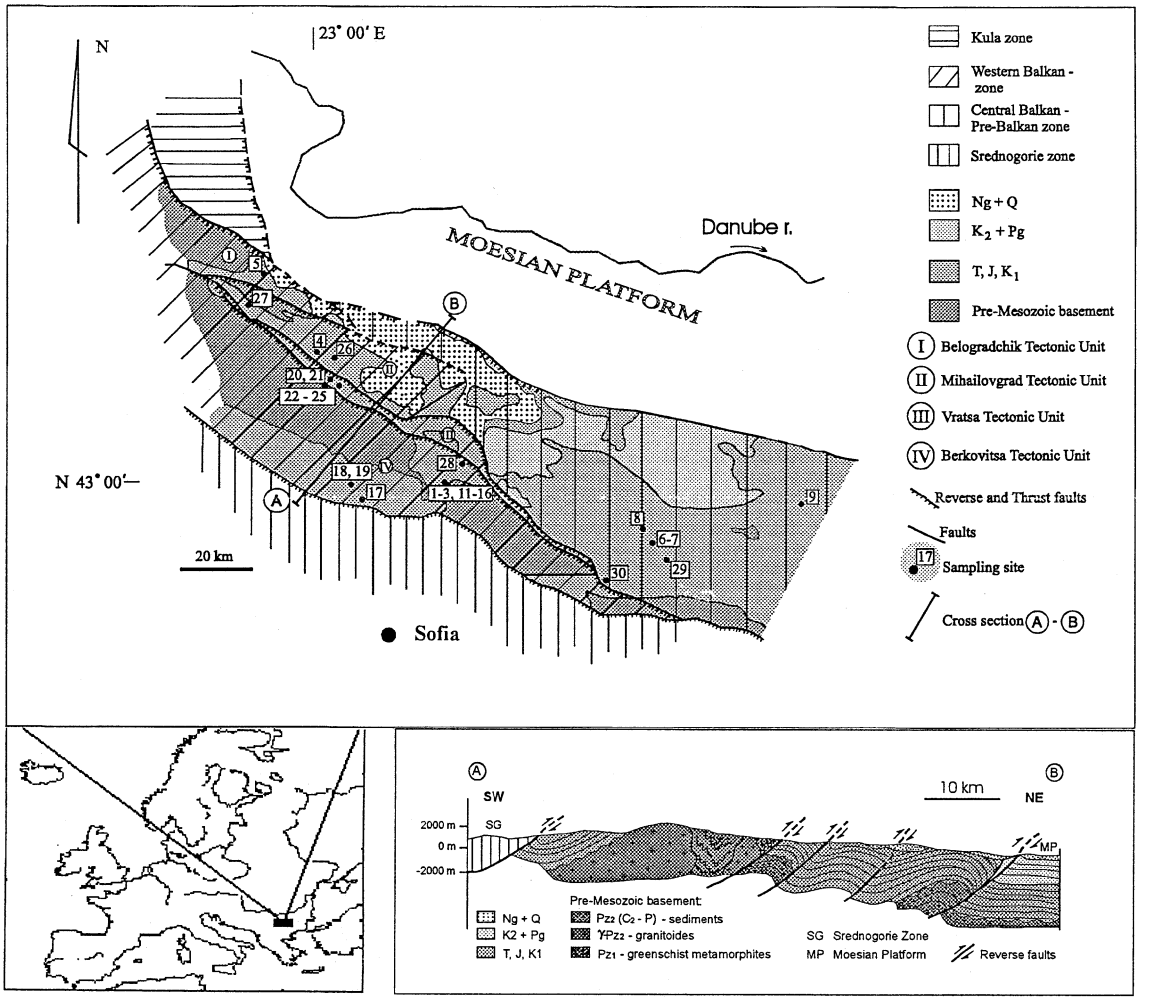


Fig. 1. Geological map with sampling sites (see site description in Table 1), simplified geological section (A–B) across West Balkan zone and sketch map of the studied locality in Europe. Q, Quaternary; Ng, Neogene; Pg, Paleogene; K_2, Upper Cretaceous; K_1, Lower Cretaceous; J, Jurassic; T, Triassic; Pz_2 (C_2-P), Permo-Carboniferous; γ-Pz_2, Upper Paleozoic intrusions; Pz_1, Lower Paleozoic.

Previous paleomagnetic results obtained in northwestern Bulgaria (Nozharov et al., 1980; Kruczyk et al., 1988; Surmont et al., 1991; Haubold and Mauritsch, 1998; Muttoni et al., 2000) yield relatively scattered directions, perhaps due to the complicated structural evolution in this area. It cannot be ruled out, however, that this scatter is related to magnetic overprinting, because many of the studies lacked rigorous paleomagnetic tests. Because paleomagnetic data can provide key tectonic information only as much as the paleomagnetic results can be considered reliable, the present work was aimed at obtaining from Balkan and Pre-Balkan areas detailed data to which stringent tests can be used to determine if remagnetization occurred in this area.

2. Geological setting

The Moesian platform and west Balkan region (Fig. 1) have similar early Mesozoic evolution (Ivanov, 1988a,b). In most cases, Permo-Triassic sediments cover transgressively and discordantly variegated Variscan basement (Tronkov, 1973). The series

Table 1
Site descriptions: location, age, number of samples (N), mean direction (D, I) before and after dip correction and corresponding Fisher's (1953) parameters (k, α_{95}) are given

Site no.	Site (lat/long)	Age	N	Before				After intermediate and				Total dip correction			
				D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)
ChRM A															
17	Dolno Butchino (42°57′27″N, 23°09′39″E)	Lower Triassic	5	36.9	27.3	24	12.7	–	–	–	–	26.9	16.1	24	12.7
22	Chiprovtzi monastery (43°24′55″N, 22°55′56″E)	Lower Triassic	6	11.6	37.4	25	11.5	–	–	–	–	37.6	17.4	25	11.5
18	Gintzi (43°03′53″N, 23°08′02″E)	Lower Triassic	3	39.1	36.1	29	14.9	–	–	–	–	39.1	36.1	29	14.9
12	Lakatnik (43°06′54″N, 23°25′42″E)	Lower Triassic	7	42.6	36.6	81	5.9	–	–	–	–	34.4	22.5	81	5.9
14	Lakatnik (43°06′18″N, 23°23′40″E)	Lower Triassic	8	29.1	41.7	46	7.3	–	–	–	–	28.0	32.8	46	7.3
15	Lakatnik (43°05′42″N, 23°22′18″E)	Lower Triassic	6	34.9	41.5	44	8.7	–	–	–	–	31.8	36.9	44	8.7
20	2 km S of Meljane (43°21′18″N, 23°01′E)	Lower Triassic	4	38.7	42.9	28	13.3	–	–	–	–	30.9	28.4	28	13.3
Mean			7 sites	33.4	38.0	68	6.4	–	–	–	–	32.6	27.2	72	6.2
ChRM B															
6	Glojene (42°59′35″N, 24°12′21″E)	Middle Triassic	7	19.7	59.2	199	3.8	19.2	59.0	199	3.8	10.4	57.0	199	3.8
7	Glojene (42°59′35″N, 24°12′21″E)	Middle Triassic	2	28.6	39.9	–	–	18.9	55.6	–	–	13.1	59.7	–	–
19	1.7 km S of Gintzi (43°04′20″N, 23°07′26″E)	Middle–Upper Triassic	5	331.0	56.8	17	15.1	2.6	48.7	50	8.9	346.6	58.1	28	11.3
1	Lakatnik (43°05′20″N, 23°23′28″E)	Middle–Upper Triassic	8	20.0	62.2	414	2.4	15.6	59.2	414	2.4	9.9	53.5	414	2.4
2	Lakatnik (43°05′20″N, 23°23′28″E)	Middle–Upper Triassic	7	31.3	61.3	167	4.1	21.6	57.0	167	4.1	20.7	56.5	167	4.1
3	Lakatnik (43°05′20″N, 23°23′28″E)	Middle–Upper Triassic	6	14.2	56.3	96	5.8	22.2	56.9	96	5.8	4.4	54.4	96	5.8
11	Lakatnik (43°07′40″N, 23°27′E)	Middle–Upper Triassic	7	63.9	71.9	215	3.6	25.9	55.7	215	3.6	21.7	53.4	215	3.6
13	Lakatnik (43°06′54″N, 23°25′20″E)	Middle Triassic	8	36.3	62.6	91	5.2	22.9	56.3	91	5.2	14.0	50.1	91	5.2
16	Lakatnik (43°05′42″N, 23°22′27″E)	Middle Triassic	5	22.8	49.5	103	6.2	27.6	54.9	103	6.2	20.4	46.2	103	6.2
30	Lopian (42°51′19″N, 24°04′16″E)	Middle Triassic	2	6.6	62.6	–	–	6.7	62.7	–	–	346.8	18.6	–	–
21	Melljane (43°22′05″N, 23°01′30″E)	Middle Triassic	9	20.7	54.8	79	5.2	20.7	54.8	79	5.2	20.7	54.8	79	5.2
23	Pomejdin village (43°23′40″N, 22°59′E)	Middle Triassic	8	343.8	63.1	202	3.5	19.2	54.9	202	3.5	36.4	30.9	202	3.5
24	Lopuchan monastery (43°22′46″N, 23°00′45″E)	Middle Triassic	7	41.2	56.9	354	2.89	37.2	53.1	354	2.89	28.1	40.3	354	2.8
25	Lopuchan monastery (43°22′46″N, 23°00′45″E)	Middle Triassic	6	31.1	64.1	338	3.1	24.9	57.1	322	3.2	17.7	38.0	142	4.8
26	Tseljustnitza quarry (43°27′08″N, 23°00′31″E)	Middle Triassic	5	28.0	32.0	33	10.9	25.2	56.8	33	10.9	27.5	48.0	33	10.9

(continued on next page)

Table 1 (*continued*)

Site no.	Site (lat/long)	Age	*N*	Before				After intermediate and				Total dip correction			
				D (°)	*I* (°)	*k*	α_{95} (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)
29	Teteven (42°54′52″N, 24°19′E)	Middle–Upper Triassic	6	11.3	56.2	722	2.1	22.0	55.7	722	2.1	352.6	54.3	722	2.1
28	Zgurigrad (43°11′25″N, 23°32′E)	Middle–Upper Triassic	4	215.2	31.9	242	4.5	29.6	57.0	242	4.5	22.3	54.8	242	4.5
20	2 km S of Meljane (43°21′18″N, 23°01′E)	Middle–Upper Triassic	3	41.2	73.4	39	12.9	19.2	57.2	39	12.9	18.1	58.2	39	12.9
5	Orechetz (43°39′N, 22°41′58″E)	Middle–Upper Jurassic	4	354.2	52.0	226	4.7	14.5	48.6	226	4.7	9.9	50.0	226	4.7
4	Mitrovtzi (43°21′18″N, 23°01′E)	Upper Jurassic–Lower Cretaceous	5	13.3	53.8	99	6.3	12.6	55.0	99	6.3	16.1	48.2	99	6.3
8	Glojene (43°00′25″N, 24°11′43″E)	Upper Jurassic	10	26.0	52.8	312	2.5	27.1	56.1	312	2.5	33.6	67.2	312	2.5
27	Warbovo (43°33′06″N, 22°40′E)	Upper Jurassic–Lower Cretaceous	7	175.3	−8.6	36	8.8	173.5	−55.9	36	8.8	172.4	−31.3	36	8.8
9	Lovetch (43°04′N, 24°44′E)	Lower Cretaceous	2	40.6	58.2	–	–	27.7	53.3	–	–	18.5	46.8	–	–
10	Boaza (42°56′48″N, 25°54′02″E)	Upper Cretaceous	5	14.6	56.1	17	15.1	15.8	58.0	17	15.1	352.5	49.1	17	15.1
Mean			23 sites	18.9	64.1	14	7.7	20.8	56.0	238	1.9	13.9	50.8	35	4.9

comprises terrigenious continental deposits (''Bundsandstein'') in the lower part, which gradually change to shallow marine terrigenious–carbonate sediments of Middle–Upper Triassic age having transgressive characteristics. Jurassic sediments lay unconformably on certain parts of Triassic rocks or directly above the Variscan basement. This unconformity is a result of late Triassic tectonic movements (the so-called early Cimmerian phase), related to intense extension occurring in the northern margin of the Tethys, as manifested by flat lying listric-like normal faults (Ivanov, in press). Jurassic units are represented by epicontinental shallow marine sediments that grade into moderate deep-water facies. By the Upper Jurassic, the basin had acquired a bathyal character. In the western parts of the Pre-Balkan, West Balkan and Moesian Platform, Upper Jurassic Kimmeridgian–Tithonian deposits are carbonate facies, while the sediments of the same age from Central Balkan and Central Pre-Balkan are flysch facies (Sapunov and Tchumachenko, 1983; Sapunov et al., 1988). Nowhere in the three study areas is there any stratigraphic or structural discordance, or interruption observed between the Upper Jurassic and Lower Cretaceous deposits. The lithology shows that these deposits have well-expressed bathyal zonation and were formed in an epicontinental basin. After the Aptian, interruptions in the sedimentation resulted from regional tectonics. All available data suggest that this tectonic event mainly affected the central Stara Planina, and central and eastern Pre-Balkan regions. Insufficient evidence exists to ascertain whether this event took place in the West Balkan area as well (Ivanov, 1983). The Upper Cretaceous deposits on the Moesian Platform and Pre-Balkan area which are of shallow-marine, epicontinental, neritic origin (Yolkichev, 1989), were transgressive at various times during Lower Cretaceous and older sections. Both Upper Cretaceous and Early Tertiary sediments were deformed at the end of the middle Eocene during ''Illyrian'' phase. As a result, the central and eastern parts of the Pre- and Central Balkan areas have been refolded. This late Alpine event was intense in the West Balkan area as well and represents the main structural phase there. It resulted in the formation of deeply penetrating northwest-trending shear zones. Northeast vergent thrusting along these zones has displaced and overlapped para-autochthonous slices of basement and sedimentary cover (Ivanov, 1988b). These slices form a

series of asymmetric anticlines (from SW to NE they are: Berkovitza, Vratza, Mihailovgrad, and Belogradchik; Fig. 1) that reflect more intense folding in their northern limbs (Ivanov, 1983).

3. Sampling

We performed preliminary sampling on rocks spanning a wide range of ages, locations, and structural orientations in northwestern Bulgaria (Fig. 1). Results reveal that the formations yielding the best results are of Triassic age, leading us to focus subsequent sampling on Upper Permian–Lower Triassic (''Bundsandstein'' red sandstone) and Middle–Upper Triassic (carbonates) units. In total, 8 sites (58 samples) in Lower Triassic, 19 sites (174 samples) in Middle–Upper Triassic, 1 site (14 samples) in Middle Jurassic, 4 sites (37 samples) in the Upper Jurassic, and 4 sites (34 samples) in the Lower Cretaceous rocks were obtained. Intensity of natural remanent magnetization (NRM) of the samples from 5 sites was too low to obtain reliable results, so these localities were not included in the data analysis (these include: 1 site in Lower Triassic, 1 site in Middle–Upper Triassic, 1 site in Middle Jurassic, and 2 sites in Lower Cretaceous rocks), leaving 31 sites that were retained (Fig. 1, Table 1).

4. Rock magnetism

Selected samples from all localities have been subjected to rock magnetic studies to obtain information about the main ferromagnetic carriers, their domain state, stability upon laboratory treatments, etc. The intensity of NRM for the collection studied varies widely (ranging from $\sim 10^{-6}$ to 10^{-3} A/m) even within the same formation. For example, Middle Triassic limestones from sites 6 and 2 are characterized by relatively strong NRM values (2.5–8 mA/m), while the same limestone from site 3 is weakly magnetic (NRM varies between 0.1 and 1 mA/m). Bundsandstein samples also show scattered NRM intensities of the same order as the limestone samples.

Thermomagnetic analysis of magnetic susceptibility $K(T)$ was carried out for representative samples from sandstones (Bundsandstein) and limestones. Measurements were done on an Agico KLY-2 susceptibility bridge with a high-temperature CS-2 attachment. High-temperature behavior of magnetic susceptibility of Bundsandstein samples (Fig. 2a) reveals the presence of both hematite and magnetite, with hematite occurring in much higher concentrations. Apparently, only minor chemical or phase transformations occur during laboratory heating, since cooling and the heating curves (Fig. 2a) do not differ significantly.

Thermomagnetic behavior of limestone samples reveals the predominant presence of magnetite (Fig. 2b), although a minor amount of hematite is also evident, as previously also found in the Edivetur section (Muttoni et al., 2000). There is also some evidence for minor alteration upon heating since cooling curve shows reduced susceptibility. A small back-cycle ($\sim$500–400 °C), performed during the heating cycle (Fig. 2b), reveals a weak mineralogical

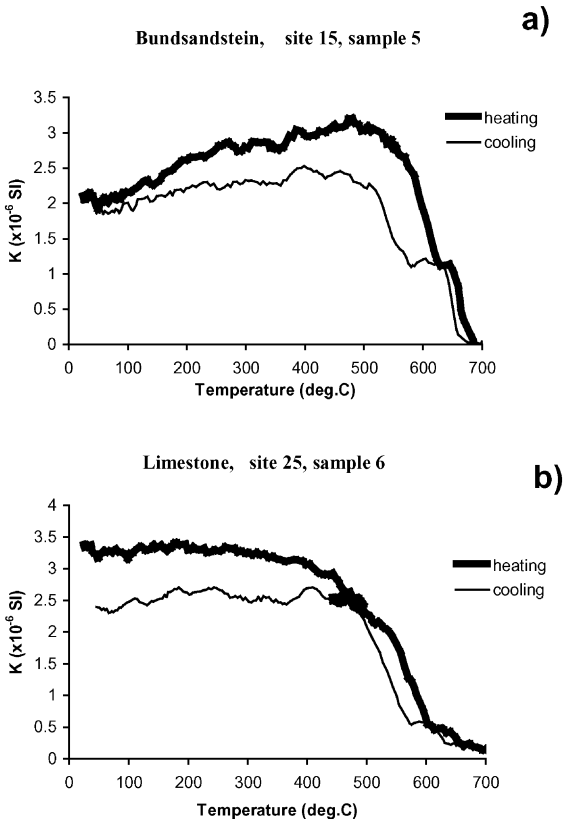


Fig. 2. Susceptibility versus temperature curves (temperature T in °C) for samples from sites 15-5 (Bundsandstein) and 25-6 (Middle–Upper Triassic).

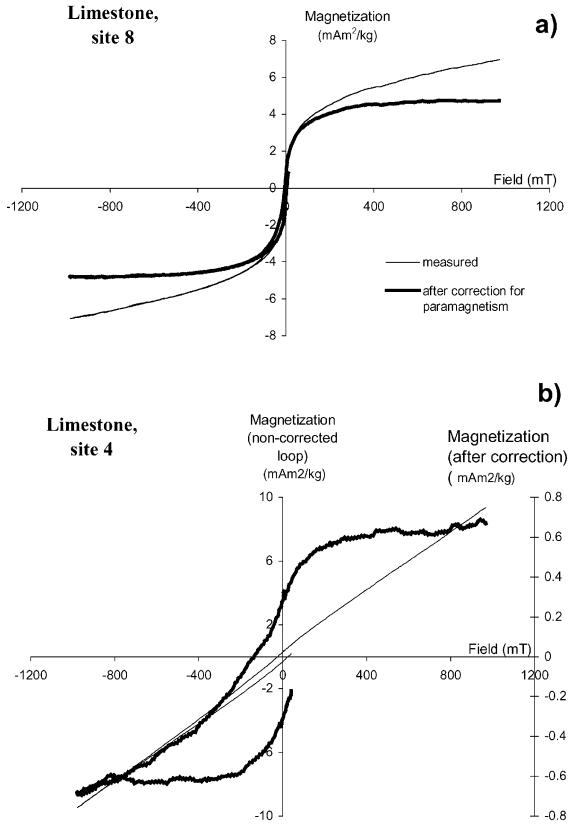


Fig. 3. Hysteresis loops (H in Tesla, J in A m^2/kg) for limestones from sites 8 and 4.

alteration by a small decrease of the susceptibility (therefore not related to formation of new magnetite). The obtained magnetite Curie temperature therefore corresponds to magnetite present before heating. In contrast, results from stepwise room-temperature measurements of magnetic susceptibility measured during thermal demagnetization reveal that, after the 400-°C temperature step, susceptibility increased rapidly, suggesting formation of new magnetite. These contradictory observations are very likely related to the fact that thermomagnetic curves were obtained in air on rock crushed to a powder and room-temperature susceptibility measurements were performed on standard specimens.

To obtain information about the coercivity and grain size of the magnetic phases, hysteresis loops were measured on bulk cylindrical samples, without magnetic separation. The experimentally obtained loops are affected by para-, dia- and ferrimagnetic fractions. Correction for para- and diamagnetic influences was done using a linear approximation in the region of the applied highest magnetic field (1 T). Para- and diamagnetic contributions to the measured hysteresis loops are very high for most of the samples (Fig. 3). The remaining part of the magnetization after correction is therefore sometimes very weak and the "ferrimagnetic" hysteresis loop strongly distorted. All "ferrimagnetic" loops for the "magnetically strongest" samples are constricted (wasp waisted), and full saturation is achieved above $\sim$0.6 T (Fig. 3). The hysteresis parameters and ratios J_{rs}/J_s and H_{cr}/H_c were also calculated for limestones (Fig. 4) pointing to the presence of various magnetic grain sizes, ranging from single domain (SD) and pseudo-single domain (PSD) to multidomain (MD) size assuming only the presence of magnetite (Day et al., 1977). Constriction could be caused by the presence, together with magnetite, either of a superparamagnetic fraction or of high-coercivity minerals such as hematite or goethite (Roberts et al., 1995; Tauxe et al., 1996). Data falling in the MD range of the Day plot (Fig. 4) are therefore probably influenced also by a hematite contribution [as evidenced also by the $K(T)$ curves], which mainly leads to higher H_{cr} and J_{rs}.

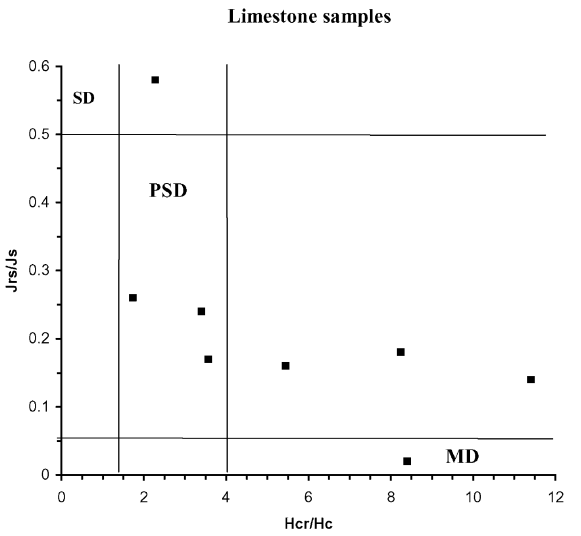


Fig. 4. Day plot showing J_{rs}/J_s as a function of H_{cr}/H_c (Day et al., 1977) for limestone samples.

5. Paleomagnetism

5.1. Analysis procedures

Before any magnetic measurements, the specimens were placed in a magnetically shielded space for over 2 weeks to reduce the viscous remanent magnetization (VRM) acquired in situ and/or after sampling. The intensity and the direction of the remanent magnetization were measured at the Institut de Physique du Globe de Paris (IPGP) Paris laboratory on 2 G cryogenic and JR-5 spinner magnetometers, and at the IPGP Saint-Maur laboratory on a JR-4 spinner magnetometer. Detailed stepwise alternating magnetic field (AF) and thermal demagnetization techniques, applied to pilot specimens, reveal that thermal treatment was more effective at isolating characteristic remanent magnetizations (ChRM).

5.2. Data analysis

Thermal demagnetization of NRM for most of the samples reveals the presence of two remanence components. The first, corresponding to unblocking temperatures below 300 °C, is close to the direction of the present geomagnetic field, and is likely of viscous origin. The ChRM of the Lower Triassic red sandstone, determined from thermal treatment up to 660 °C, is clearly mainly carried by hematite (Fig. 5a). For all other samples, however, the maximum blocking temperature of the ChRM varies between 400 and 580 °C (Fig. 5b). Two groups of magnetizations can be distinguished according to their paleomagnetic direction. Group A (moderate inclination towards the NNE, main carrier hematite) corresponds to Bundsandstein samples, while group B (mostly steeper inclination, main carrier magnetite) is found at all other sites. Six Bundsandstein samples, presenting directions different from the other Bundsandstein samples but similar to that of the group B samples, were discarded.

5.3. ChRM of group A

The fold test result is not significant (at 95% confidence) for group A (Fig. 6 and Table 1), because of shallow and relatively similar dips in all Bundsandstein sites. The polarity of the ChRM A is always normal, except for one sample. In the site where both polarities are obtained, the ChRM vectors are roughly antiparallel, showing that ChRM does not result from the superposition of different magnetization components.

The paleomagnetic pole position (Fig. 7) calculated from our data (148.1°E, 50.0°N, $K=142$, $A_{95}=4.4°$), lies close to the Lower–Middle Triassic poles (Van der Voo, 1990; Edel and Duringer, 1997; Besse and Courtillot, in press) of the European Apparent Polar Wander Path (APWP), suggesting that the ChRM A very likely reflects a primary magnetization: the lack of paleomagnetic tests, however, does not allow us to prove definitely the timing of magnetization acquisition. Provided the direction is primary, it suggests, when compared to the previous studies (e.g. Muttoni et al., 2000), that northwestern Bulgaria did not undergo significant rotation relative to stable Europe since at least the Early Triassic. Owing to the window of possible ages of deposition of the rocks with Bundsandstein facies (Upper Permian to Lower Triassic), a clockwise rotation of the studied area cannot be excluded.

Nozharov et al. (1980) report a mean paleomagnetic direction ($D=26.1°$, $I=39.3°$) for Bundsandstein sites in northern Bulgaria that is close to our ChRM A component, though slightly steeper in inclination. This slight difference may be due to the inclusion of some ChRM B data (as measured in the six discarded samples in our sites) in their estimate of the primary magnetization; their mean direction is in fact located precisely between our A ($D=32.6°$, $I=27.2°$) and B directions ($D=20.8°$, $I=56.0°$).

5.4. ChRM of group B

All the directions of group B are of normal polarity (an exception is site 27, Table 1). Although this site was discarded from subsequent considerations, its exclusion from the calculation of the group B mean does not significantly change the results. One locality of Middle–Upper Triassic age (site 26) is represented by a conglomeratic facies, with limestone clasts within a fine carbonated matrix. The magnetization direction obtained for the clasts and the matrix coincides (negative conglomerate test), showing that the ChRM B is a remagnetization. The presence of ChRM B in six Bundsandstein samples also points out this remagnetization.

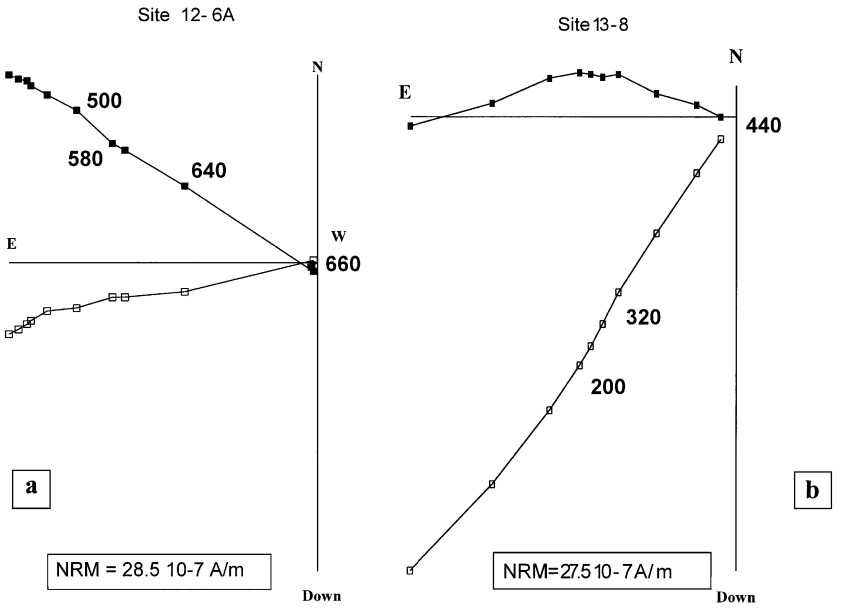


Fig. 5. Examples of orthogonal vector plots: projections on horizontal (full squares) and vertical (open squares) planes: (a) specimens 12-6A (Bundsandstein) and (b) 13-8 (Middle–Lower Triassic); temperature in °C.

A fold test was also performed using the mean ChRM B direction for each site (Table 1). Some of the group B magnetizations, which were obtained from a very highly deformed zone restricted to part of the Middle–Upper Triassic site 7 (within a narrow fault zone), were eliminated from the analysis because we were not able to perform a reliable dip correction to them (due to the presence of several-fold axes with different dips). During progressive unfolding (Fig. 8), the precision parameter k (Fisher, 1953) is maximum value at 60% unfolding and the distribution of the directions appears to be closest to Fisherian (minimum r value, Henry and Le Goff, 1994) at 70% unfolding. The ChRM B is therefore a syntectonic magnetization that was acquired sometime between the beginning of the first deformation and the end of

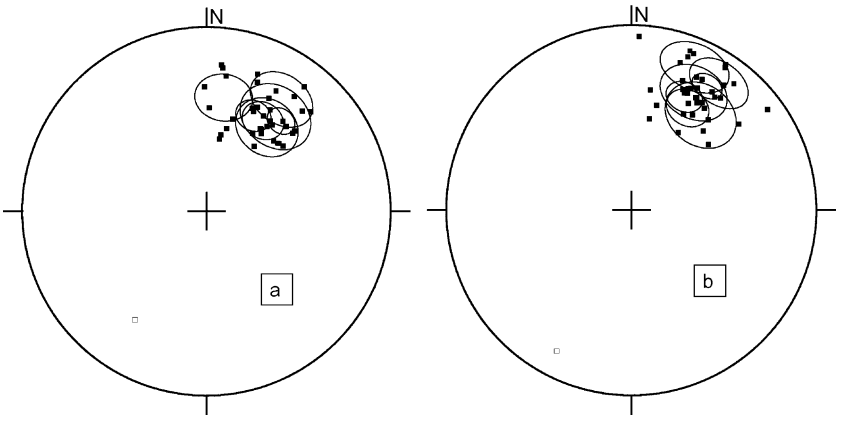


Fig. 6. ChRM A directions (squares) with 95% confidence zones for each site: stereographic projection, the full (empty) symbols corresponding to the lower (upper) hemisphere before (a) and after (b) dip correction. Confidence zones in the lower hemisphere.

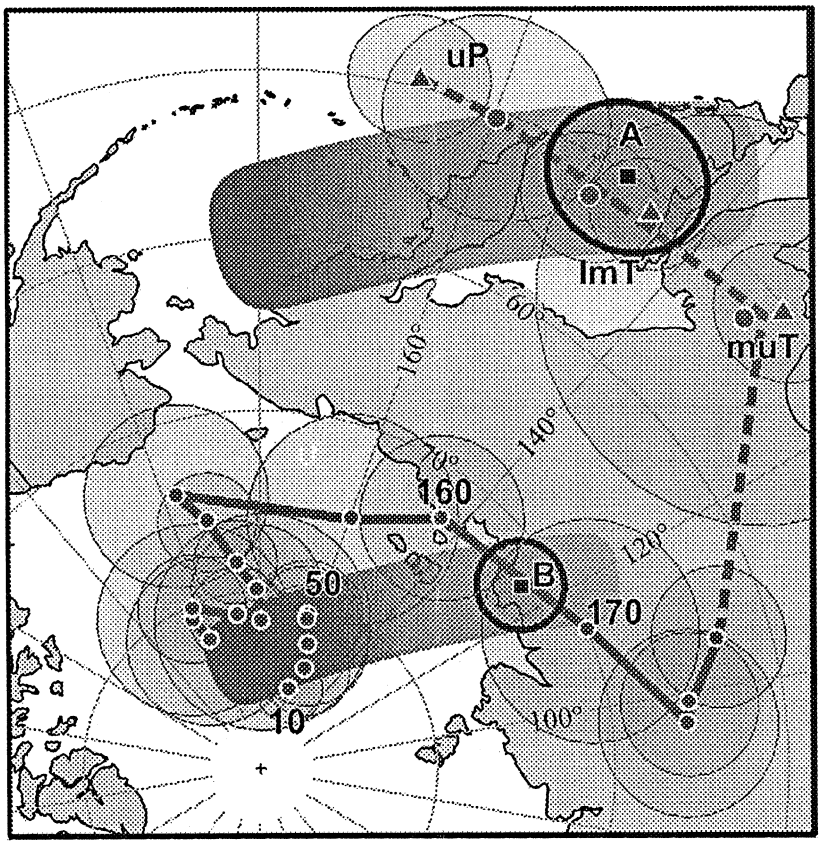


Fig. 7. Paleomagnetic poles for stable Europe (small circles — Besse and Courtillot, 2001; large circles — Van der Voo, 1990; triangles — Edel and Duringer, 1997) and poles corresponding (squares) to ChRMs A and B. Age in Ma or by geological period (muT — Middle Upper Triassic, lmT — Lower–Middle Triassic, and uP — Upper Permian). The broad gray arcs of circle correspond to poles A and B assuming a rotation of the study area.

the last deformation (i.e. possibly between two tectonic phases). For such syntectonic magnetization, involving widely distributed sites, the assumption that all sites simultaneously share the same unfolding percentage is unrealistic. Surmont et al. (1990) and Shipunov (1997) proposed a different method, which yields, at each site, the direction of the syntectonic magnetization and the folding percentage. Their method makes use of the fact that, during tilting, the magnetization direction moves along a small circle (great circle when the magnetization direction is perpendicular to the fold axis). The intersection of circles from different sites represents the unique possible shared magnetization direction, and therefore the syntectonic remagnetization. To determine this intersection (Figs. 9 and 10), a least squares technique was applied using the iterative method of Surmont et al. (1990). This yielded a well-defined mean direction for the remagnetization ($D=20.8°$, $I=56.0°$, $k=238$, $\alpha_{95}=1.9°$) as well as the dip at the time of acquisition of the remagnetization at each site (Fig. 11).

Despite the lack of significant paleomagnetic tests (e.g. fold and conglomerate tests), some previous paleomagnetic studies (Kruczyk et al., 1988; Surmont et al., 1991) were interpreted as yielding primary magnetizations. Muttoni et al. (2000) argue for primary magnetization for another Triassic section at Edivetur (Belogradchik region) on the basis of paleomagnetic pole position analysis. Their mean paleomagnetic direction ($D=25.5°$, $I=59.7°$ before dip correction,

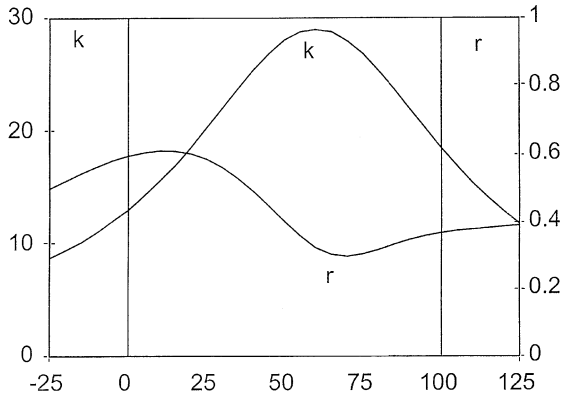


Fig. 8. Variation of precision parameter k and of correlation coefficient r (Henry and Le Goff, 1994) during progressive unfolding from −25% to 125% for ChRM B.

Muttoni et al., 2000) and those from two Callovian sites in the study of Kruczyk et al. (1988), however, are very close to our remagnetization B direction, suggesting they are not of primary origin. Unfortunately, the paleomagnetic pole positions only crudely constrain the age of magnetization acquisition. It is possible, therefore, that the dominant reversed polarity in the Edivetur sites (only six samples carry normal polarity), which agrees with the dominant polarity during the deposition of the series (Muttoni et al., 2000), indicates that the magnetization in the Edivetur sites, like that for the Callovian sites of Kruczyk et al. (1988), is different from that of the other sites (note that only two of the six normal directions could be roughly correlated with other Triassic sections). Application of the Surmont et al. (1990)–Shipunov (1997) method, however, to all the other data of Kruczyk et al. (1988) and Surmont et al. (1991) gives a best intersection similar to that obtained with our data, thus confirming that their ChRM is likely the same as our syntectonic remagnetization B. The mean direction obtained using all the usable data (from Kruczyk et al., 1988; Surmont et al., 1991; and our work) is $D=22.3°$, $I=56.5°$, $k=254$, $\alpha_{95}=1.4°$.

The corresponding paleomagnetic pole (125.2°E, 72.0°N, $K=134$, $A_{95}=1.9°$) is compared with the European APWP (Besse and Courtillot, in press) in Fig. 7. It corresponds to about 165 Ma (Dogger) poles. However, because the ChRM B has been found also in Lower Cretaceous formations, such a Dogger age is not possible. Therefore, we considered the possibility that the area was subjected to vertical axis rotations (Fig. 7). It appears that a Tertiary (Eocene) age would be possible if we suppose that the study area underwent a clockwise rotation of the order of $14° \pm 6°$. Such an Eocene age is also supported by the fact that our paleomagnetic direction is very close to that, unfortunately imprecise ($D=22.6°$, $I=53.6°$, $\alpha_{95}=15.8°$), obtained by Haubold et al. (1998) for Paleogene rocks from the Moesian platform. An additional constraint on the age of the ChRM B is given by the "syntectonic" character of the magnetization. At the time of remagnetization, dips exceeded 20° at some sites (Fig. 11), implying that remagnetization accompanied relatively significant tectonic movements. This eliminates Mesozoic ages for the magnetic overprinting. It is thus very probable that the remagnetization occurred during the first stages of the Illyrian deformation (end of Middle Eocene) rather than during or after the possible so-called Laramian phase at the end of the Early Paleocene, that is found only locally in the Srednogorie area (Nachev, 1980). The obtained near-zero values of tilting at the time of remanence acquisition of the B component for sites 1, 2, 20, and 28 (Fig. 11) do not imply that it can be

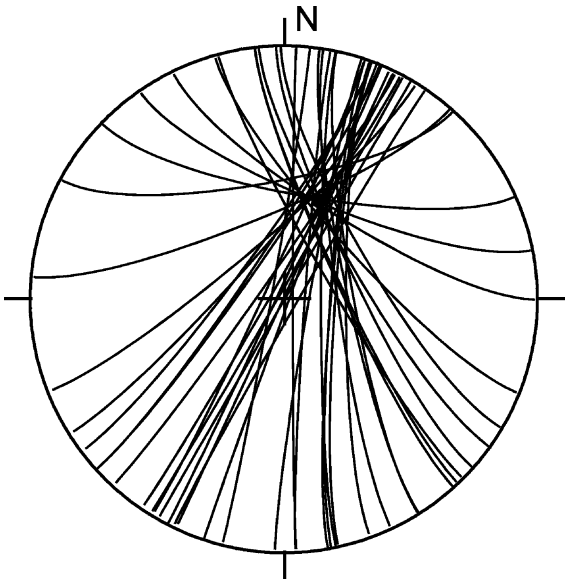


Fig. 9. Small circles each including the possible orientation of the mean ChRM B for each site (Surmont et al., 1990; Shipunov, 1997). Stereographic projection in the lower hemisphere.

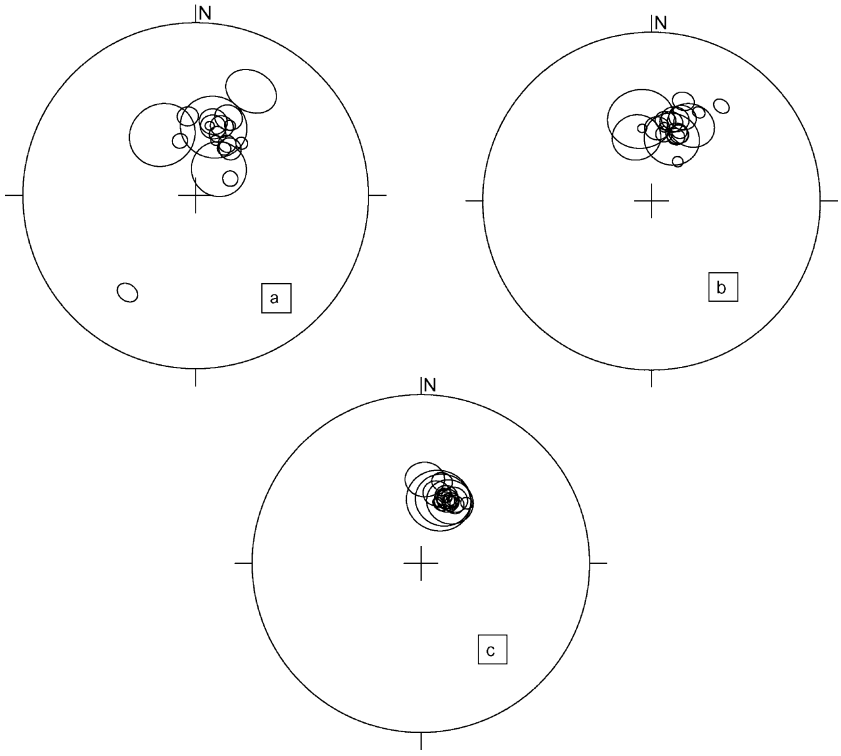


Fig. 10. ChRM B confidence zone for each site: stereographic projection in the lower hemisphere before (a) and after (b) dip correction, and at optimal unfolding (c).

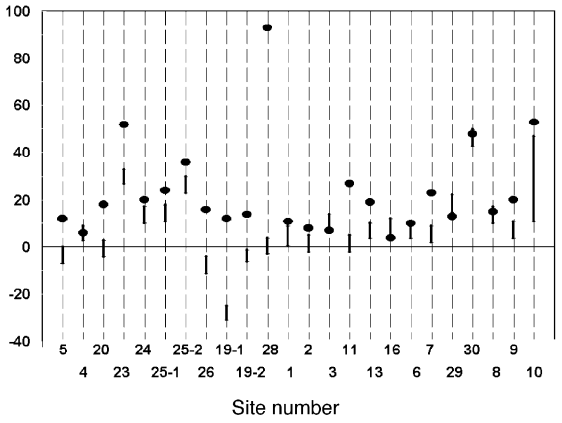


Fig. 11. Dip at the time of remagnetization at the different sites, (vertical line — uncertainty has been estimated using bootstrap method — Efron, 1982; Efron and Tibshirani, 1986; Henry, 2000) compared to present dip (full ellipse). Negative values indicate dip in the opposite direction of the present dip. Sites subdivided in two parts (sites 25:25-1 and 25-2; sites 19:19-1 and 19-2) contain samples with two different present dip values. For site descriptions, see Fig. 1.

primary because the sites referred are of Middle–Upper Triassic age (Table 1), while the calculated pole position corresponds to Dogger, accepting the hypothesis for the absence of rotation.

The maximum blocking temperature of the ChRM is too high to be related to a thermal event and the remagnetization is therefore considered to be of chemical origin. The fact that only normal polarity was found does not necessarily indicate this remagnetization was acquired during a single period of normal polarity during the Eocene. Because the remagnetization is of chemical origin, thermal or alternating field demagnetization in general cannot separate different components of magnetization. Only the polarity associated with the largest number of magnetic grains can be obtained. Moreover, during grain growth, magnetic moment is blocked as soon as the blocking volume is reached, and up to relatively large MD sizes. Consequently, the normal polarity of the ChRM B only indicates that more than 50% of the grains reached

their blocking volume during a period of normal polarity of the geomagnetic field. Thus, the duration of the acquisition of the ChRM B could be longer than a single normal polarity chron (all of which are relatively short during the Eocene).

6. Geological implications

Clearly, this remagnetization, which is found over a wide area, cannot be due to a local event. Two phenomena (fluid migration—e.g. Sangster, 1986; Symons et al., 1996—or illitization—e.g. Katz et al., 1998) could be responsible for the crystallization of magnetite or transformation of original minerals to magnetite. The presence of the same remagnetization at several stratigraphic levels and over a wide area argues for fluid migration, since illitization probably would not occur in all the places and at the same time.

It is suggested by the ChRM B that northwestern Bulgaria underwent clockwise rotation relative to staple Europe since the beginning of the Middle Eocene tectonics. Such a rotation is also compatible with the pole position of the ChRM A. The study area therefore should present an intermediate rotation between that observed in the Banat–Apuseni area in Romania (Surmont et al., 1990; Patrascu et al., 1990, 1992) and stable Europe. Considering the structural map (Fig. 1), thrust faults in the Balkan zones appear more important in the western part than in the eastern, possibly indicating that the Moesian platform was less (or not at all) affected by the rotation observed in the Balkan zones.

The mean ChRM A directions for each site are relatively similar arguing for lack of significant relative rotations, though owing to the relatively large uncertainty of these directions (Table 1 and Fig. 6), we cannot exclude the existence of small local movements. The coherence of small circles (Fig. 9) from ChRM B also suggests a lack of significant relative rotations since Middle Eocene.

Similar magnetic overprints of normal polarity (believed to be either Tertiary or Late Cretaceous in age) have also been recovered in the Srednogorie area in the southern Bulgaria (Kruczyk et al., 1990). This region was probably deformed after acquisition of the remagnetization, since the observed inclinations are relatively high (64°) with respect to the expected Late Cretaceous inclination (approximately 50°). If the ChRM is of Cretaceous age, the Srednogorie direction must have been rotated by regional northward tilting of about 15°. If the age is Tertiary, the Srednogorie direction must have been affected by later northward tilting of about 10° and counterclockwise rotation (10–20°) relative to the Balkan area. In either case, the boundary between Srednogorie and Balkan areas appears therefore to be an important structure separating different tectonic domains.

At the time of the remagnetization, the dip (Fig. 11) was for most of the sites shallow (and less steep than the present dip) and towards the NNE (for the sites 26 and 19, the NW and NNE dips were in the direction opposite to that of the present dips), although relatively high dips towards NNE were determined in some areas for the Eocene. This suggests that stronger uplift occurred within the Srednogorie zone than within the Moesian platform.

The fact, that the remagnetization was acquired before complete folding at most of the sites (Fig. 11), constrains the age of the remagnetization to the beginning of the Illyrian phase. However, like in the Betics (Villalain et al., 1994, 1996), some sites show pre-, syn- or post-folding remagnetization (Fig. 11). The degree of dip at the time of remagnetization reveals the relative timing of deformation and remagnetization. At sites just north of the thrust faults in the West Balkan area (sites 4 and 30—Fig. 1), deformation had completed before remagnetization acquisition. In contrast, sites 28 and 5 (Fig. 1), located far from such thrust faults, were not yet affected by deformation by the time of magnetic overprinting. At site 26, which is presently on the southern limb of an anticline also just south of a thrust fault, the dip was moderate at the time of remagnetization and in the opposite direction, showing that this anticline very likely had not yet formed by the time of remagnetization. For most of the sites in the middle of the zone, delimited by the thrust faults, the remagnetization was acquired during an intermediate stage of folding. This indicates therefore there was either diachronous remagnetizations or diachronous foldings.

To assume diachronous remagnetization and the same age for all the Illyrian folding implies that the remagnetization began from the northern border in each unit (before folding), and progressively reached the southern part (during folding in the middle of each unit and after folding at their southern borders).

This would suggest that fluids migrated from the low-lying Moesian platform towards the uplifted and deformed Srednogorie zone. Such an assumption, however, is un-realistic, because fluid migration, due either to gravity effect (see, e.g., Garven and Freeze, 1984a,b) or to tectonic compression (Oliver, 1986), would proceed from south to north. On the other hand, synfolding remagnetization has been obtained in the middle of several units: for example, site 19 in the first unit from the south (Berkovitza unit), 23 in the second unit, and 9 in the Central Pre-Balkan (Fig. 1). To assume the same age for the folding therefore implies the same age of remagnetization in the middle of the different units, which are not at the same latitude. The fact that remagnetization was contemporaneous in these different units suggests that fluid migration started roughly at the same time in these units. This argues for a local origin of the fluids (thus fluids expelled from deformed zones between and within each unit—Oliver, 1986). Because these units are not very large, any possible differences in age of remagnetization within them are likely small.

Folding therefore was not synchronous in the whole area. Deformation began, in several areas, which later became the southern border of each unit (i.e. the levels just below the present discontinuities). It progressively reached areas farther to the north. The last affected levels are those just above the next discontinuity. The evolution of the deformation in this area was very different on either side of the present discontinuity.

Thus, the deformation started along more or less parallel zones that possibly correspond to previous (sedimentary or tectonic) discontinuities. Around each "discontinuity", the overlying southern part acted as a rigid block, and the "folding" front advanced northward in the underlying northern part. Progressively, the whole unit was affected by the folding.

7. Conclusion

The characteristic component of magnetization (ChRM A) observed in Upper Permian–Lower Triassic sediments in northwestern Bulgaria is likely a primary direction. The orientation of a secondary component (ChRM B) is believed to reflect a clockwise rotation of this area relative to stable Europe after the beginning of the main Middle Eocene tectonic phase. The ChRM B provides a unique picture of the geological structures at an intermediate stage of deformation, and reveals that Eocene deformation began in the south and proceeded northwards.

Acknowledgements

This work was supported by the Bulgarian Academy of Sciences, the French CNRS, and the Péri-Téthys program. We are very grateful to J. Glen for discussion and help to improve the manuscript and to reviewers R. Van der Voo and G. Muttoni for constructive comments.

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